29 April 2024 10:54

Properties of DFT

- **Circular shift**

$\{x(n)\}$ $\xleftrightarrow{N\text{ pt}}$ $\{X(k)\}$ $k=0,\dots,N-1$

N length

$\Downarrow$

circular shift by N_0 units

$$x_1(n) = x((n-N_0) \bmod N)$$

$$\{x_1(n)\} \xRightarrow{N\text{ DFT}} X_1(k) = e^{-j\frac{2\pi}{N}kN_0} X(k)$$

- **Symmetry.**

$$\{x(n)\} \xleftrightarrow{N\text{ pt}} \{X(k)\} \qquad k=0,\dots,N-1$$

$$\text{Conjgt } \{x^*(n)\} \xleftrightarrow{N\text{ pt}} \{X^*(-k \bmod N)\}$$

$k=0,\dots,N-1$

DFT coeff. array.

$$\{x(n)\} \xleftrightarrow{N\text{pt}} [X(0), X(1), \dots, X(N-1)]$$

$\uparrow$ $k=0$

$$\{x(n)\} \xleftrightarrow{DTFT} X(\omega)$$

$$\{x^*(n)\} \leftrightarrow X^*(-\omega)$$

$$\{x^*(n)\} \xleftrightarrow{N\text{ pt DFT}} [X^*(0)\ X^*(N-1)\ X^*(N-2)\ \dots\ X^*(1)]$$

$$\{x^*(n)\} \overset{\text{N pt DFT}}{\Longleftrightarrow} [X^*(0), X^*(N-1), X^*(N-2), \ldots, X^*(1)]$$

$$\uparrow \quad k=0$$

If $\{x(n)\}$ is real valued,

Conjugate Symmetry of DFT

$$\begin{cases} X(k) = X^*(-k \bmod N) \\ X(0) = X^*(0) \quad \text{(real)} \\ X(1) = X^*(N-1) \\ \vdots \end{cases}$$

Circular Convolution

$$\{x_1(n)\} = \{x_1(0), \ldots, x_1(N-1)\}$$

$$\uparrow \quad n=0$$

$$\{x_2(n)\} = \{x_2(0), \ldots, x_2(N-1)\}$$

$$\uparrow \quad n=0$$

$$\{x_1(n)\} \overset{\text{N pt DFT}}{\longleftrightarrow} \{X_1(k)\} \quad k = 0, \ldots, N-1$$

$\xrightarrow{\text{DFT}}$ $\quad k = 0, \cdots, N-1$

$\{x_2(n)\} \xleftrightarrow{N\text{ pt}} \{X_2(k)\} \quad k = 0, \cdots, N-1$

$$Y(k) = X_1(k) \cdot X_2(k)$$

$k = 0, \cdots, N-1$

$$\{Y(k)\} \xrightarrow[N\text{ pt DFT}]{\text{inverse}} \{y(n)\}$$

$n = 0, \cdots, N-1$

$$\{y(n)\} = \{x_1(n)\} \circledast_N \{x_2(n)\}$$

N pt circular convolution

$$y(n) = \sum_{\ell=0}^{N-1} x_1(\ell)\, x_2((n-\ell) \bmod N)$$

Parseval's Theorem

$$\sum_{n=0}^{N-1} |x(n)|^2 = \frac{1}{N} \sum_{k=0}^{N-1} |X(k)|^2$$

Circular Convolution

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Circular Convolution

$x(n) = \{x_1(0), x_1(1), x_1(2), x_1(3)\}$

$h(n) = \{h(0), h(1), h(2), h(3)\}$

$\{y_c(n)\} = \{x(n)\} \circledast_4 \{h(n)\}$ — N=4

will be also length 4

$$y_c(n) = \sum_{m=0}^{N-1} x(m)\, h(n-m \bmod N)$$

$x(m)$

x(0)	x(1)	x(2)	x(3)

m

$h(m)$

h(0)	h(1)	h(2)	h(3)

m

$h(-m \bmod N)$ circular time reversed

h(0)	h(3)	h(2)	h(1)

m

$y_c(0) = x(0)h(0) + x(1)h(3) + x(2)h(2) + x(3)h(1)$

$h(1-m \bmod N)$

h(1)	h(0)	h(3)	h(2)

m

$y_c(1) = x(0)h(1) + x(1)h(0) + x(2)h(3) + x(3)h(2)$

Similarly

$y_c(2) = x(0)h(2) + x(1)h(1) + x(2)h(0) + x(3)h(3)$,

$y_c(3) = x(0)h(3) + x(1)h(2) + x(2)h(1) + x(3)h(0)$

$\downarrow$

$y_c(n) = \{y_c(0), y_c(1), y_c(2), y_c(3)\}$

length 4.

Linear Convolution

regular convolution

$\{y(n)\} = \{x(n)\} * \{h(n)\}$

$$y(n) = \sum_m x(m)\, h(n-m)$$

$x(m)$ $\downarrow$

x(0)	x(1)	x(2)	x(3)

$h(-m)$ $\downarrow$

h(3)	h(2)	h(1)	h(0)

$y(0) = x(0)h(0)$

$h(1-m)$ $\downarrow$

h(3)	h(2)	h(1)	h(0)

$y(1) = x(0)h(1) + x(1)h(0)$

similarly, $y(2) = x(0)h(2) + x(1)h(1) + x(2)h(0)$

$\vdots$

$y(5) = x(2)h(3) + x(3)h(2)$

$y(6) = x(3)h(3)$

$\downarrow$

$y(n) = \{y(0), y(1), \ldots, y(6)\}$

length 7.

Circular Convolution with sufficient zero Padding becomes linear Convolution.

Previous example

$x(n) = \{ x(0), x(1), x(2), x(3), 0, 0, 0 \}$

$h(n) = \{ h(0), h(1), h(2), h(3), 0, 0, 0 \}$

let $y_z(n) = \{x(n)\} \circledast_7 \{h(n)\}$

↓ length 7

$x(m)$ ↓

x(0)	x(1)	x(2)	x(3)	0	0	0

$h(m)$ ↓

h(0)	h(1)	h(2)	h(3)	0	0	0

$h(-m \bmod 7)$ ↓

h(0)	0	0	0	h(3)	h(2)	h(1)

$y_z(0) = x(0)\,h(0)$

$h(1-m \bmod 7)$ ↓

h(1)	h(0)	0	0	0	h(3)	h(2)

$y_z(2) = x(0)\,h(1) + x(1)\,h(0)$

and so on

Now $y_z(n) = y(n)$ (lin. conv. output)

General Result:

$\{x(n)\} \rightarrow$ length N_x

$\{h(n)\} \rightarrow$ length N_h

lin. conv. output $N_y = N_x + N_h - 1$

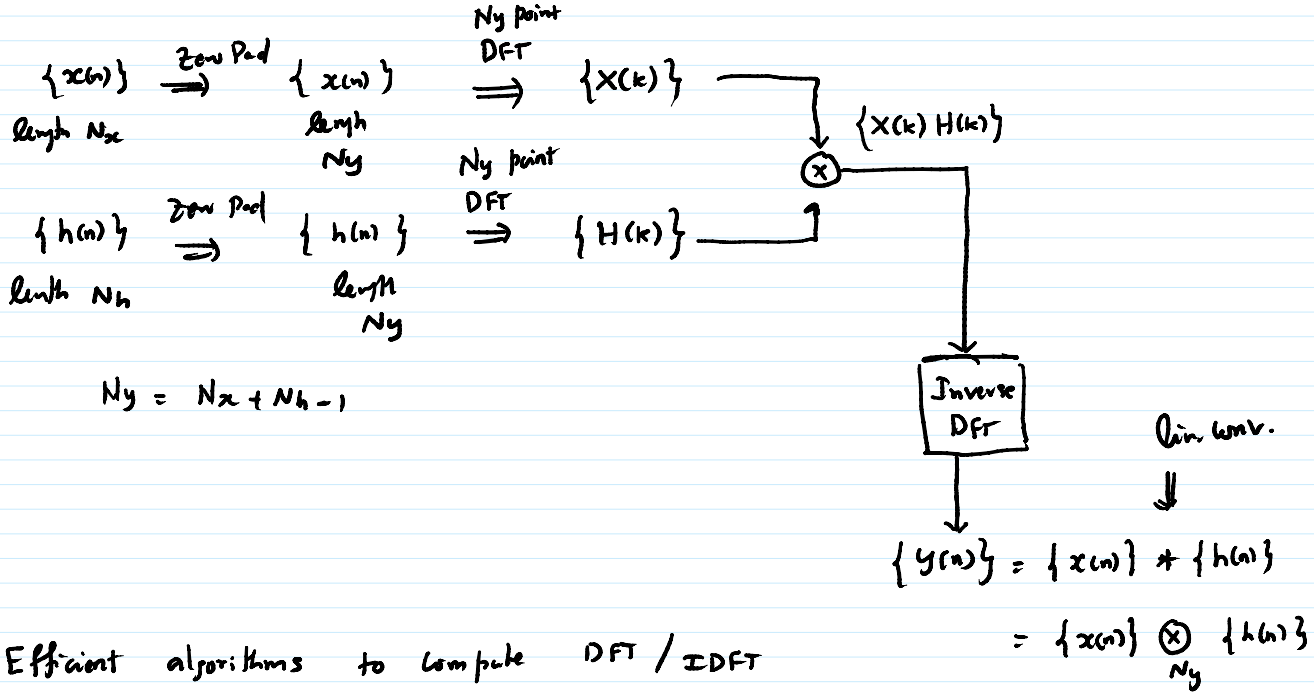
$\{x(n)\}$ ⊛ $\{h(n)\}$ = $\{y_c(n)\} = \{y(n)\}$

$\{x(n)\}$: zero pad $(N_y - N_x)$ zeros; ⊛ N_y; $\{h(n)\}$: zero pad $(N_y - N_h)$ zeros

Circ. conv. output = lin. conv. output.

Implementing Filtering Operation in Fourier domain

Using DFT & Circular Conv. Property.



Efficient algorithms to compute DFT / IDFT

This Fourier domain filtering method has computational advantages over implementing Convolution directly in time domain